Feature Extraction

Robots in Context

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Content

- Feature extraction in laser scans
  - Features
  - Line Fitting
  - Line Extraction Algorithms
- Introduction to Localization



Feature detection – why?

- Determine relation to environment
- Two basic approaches
  - From raw data
  - From higher level structures

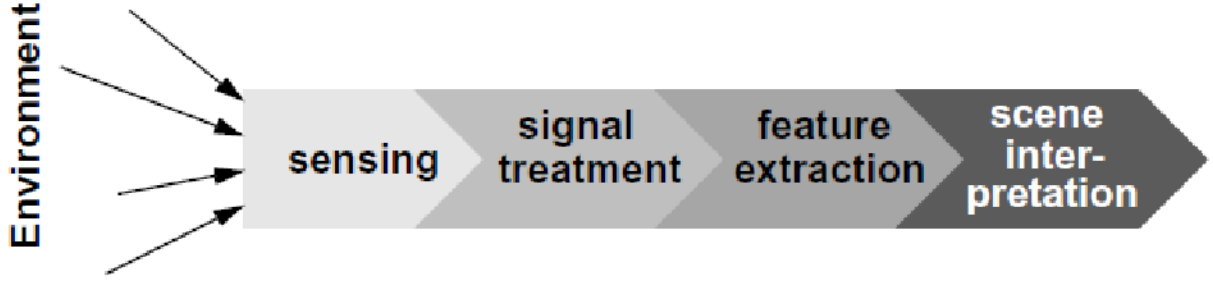


Figure 4.34
The perceptual pipeline: from sensor readings to knowledge models.

Feature detection – what?

- Good features are easily detected and robust
- Abstraction trade-off
  - Low level – lines, circles, polygons..
  - High level – doors, tables..
- Markers



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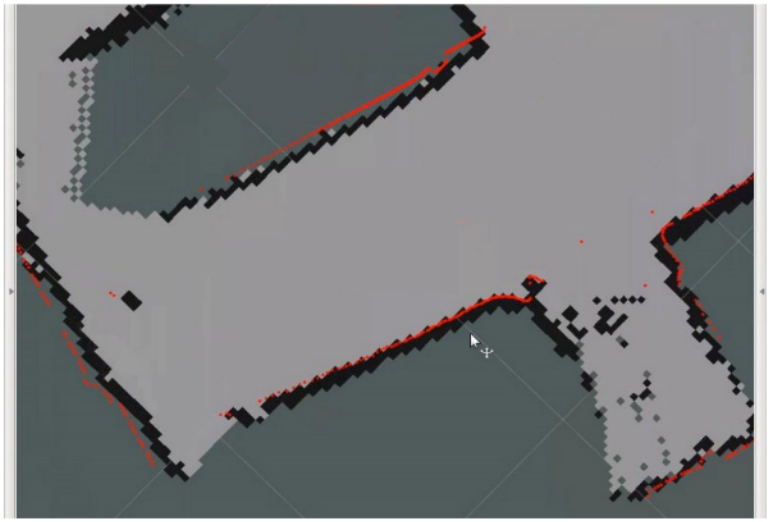


Line Features

- Lines are easily extracted
- They are very common in in-door and urban environments
- Easily combined into more complex features
  - Corner, Door ways..



Line Feature Extraction



- Input
  - A scan, or a list of points
- What is needed
  - Ability to determine which points belong to a line
  - Ability to fit a line to the points.

Line Representations

- Representation of a line:

$$x\cos(\alpha) + y\sin(\alpha) = r$$

  - α is the angle between x-axis and the normal
  - r is the distance from the origin to the closest point on the line
- Why not use y=ax+b?



Line Fitting

- A given point (x, y) can be expressed as

$$x = \rho \cos(\theta),\ y = \rho \sin(\theta)$$

- The line consists of all points where

$$x \cos(\alpha) + y \sin(\alpha) = r$$

- For x, y to be on the line:

$$\rho \cos(\theta) \cos(\alpha) + \rho \sin(\theta) \sin(\alpha) = r$$

$$\Leftrightarrow \rho \cos(\theta - \alpha) - r = 0$$

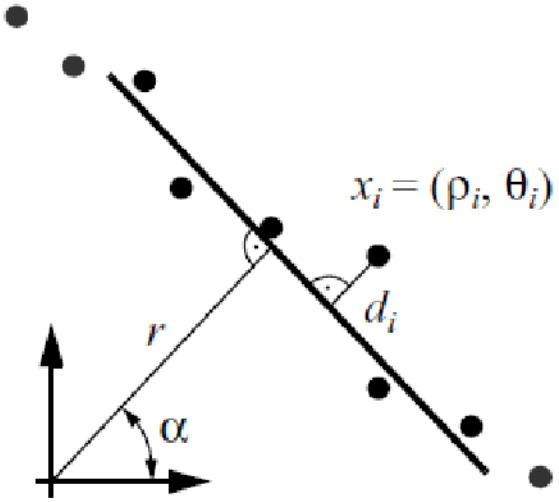


Least Square Line fitting

Assuming there is noise on the measurements

$$\rho\cos(\theta-\alpha)-r=d$$

The least square fit over N measurements: $\min_{\alpha,r} S = \min_{\alpha,r} \sum_i^N d_i^2$



We can include a weight in case not all points should have similar importance

$$S = \sum_i^N w_i\, d_i^{\ 2}$$

Least Square Line fitting

- Objective $S = \sum w_i d_i^2 = \sum w_i (\rho_i \cos(\theta_i - \alpha) - r)^2 .$

- Optimum where?



Least Square Line fitting

- Objective $S = \sum w_i d_i^2 = \sum w_i (\rho_i \cos(\theta_i - \alpha) - r)^2 .$
- Optimum where $\frac{\partial S}{\partial \alpha} = 0 \qquad \frac{\partial S}{\partial r} = 0 .$
- Solution $\alpha = \frac{1}{2}\text{atan}\left(\frac{\sum w_i \rho_i^2 \sin 2\theta_i - \frac{2}{\sum w_i} \sum\sum w_i w_j \rho_i \rho_j \cos\theta_i \sin\theta_j}{\sum w_i \rho_i^2 \cos 2\theta_i - \frac{1}{\sum w_i} \sum\sum w_i w_j \rho_i \rho_j \cos(\theta_i + \theta_j)} \right)$

$$r = \frac{\sum w_i \rho_i \cos(\theta_i - \alpha)}{\sum w_i}$$

Least Square Line Fitting - Pitfalls

- The books assumes the best fit when

$$\frac{\partial S}{\partial \alpha} = 0 \qquad \frac{\partial S}{\partial r} = 0 \,.$$

- This can also be the worst fit, hence the line perpendicular.
- To fix it
  - Set $\alpha = \alpha + \frac{\pi}{2}$ and recompute *r*.
- How to detect it
  - Option 1: Check the higher order derivative to see if it is a local minimum or maximum.
  - Option 2: Compute the residual for both cases and choose the one with the smallest residual.
  - Option 3: Use prior knowledge about the expected direction of the line
- Other pitfall
  - The algorithm may generate negative values of *r*. In this case set $\alpha = \alpha + \pi$ and change the sign of *r*.



Line Fitting

- We now know the least square solution. Do this mean that we are done?



Line Fitting

- We now know the least square solution. Do this mean that we are done?
  - No. We still do not know which points are to be associated with the line.



Line Extraction Algorithms

- Algorithm from paper
  - Split-and-merge
  - Incremental
  - RANSAC
  - Hough transform

- Will not consider
  - Line regression
  - Expectation-Maximization



Split and merge

Algorithm 1: *Split-and-Merge*

1 Initial: set s_1 consists of N points. Put s_1 in a list $\mathcal{L}$
2 Fit a line to the next set s_i in $\mathcal{L}$
3 Detect point P with maximum distance d_P to the line
4 If d_P is less than a threshold, continue (go to 2)
5 Otherwise, split s_i at P into s_{i1} and s_{i2}, replace s_i in $\mathcal{L}$ by s_{i1} and s_{i2}, continue (go to 2)
6 When all sets (segments) in $\mathcal{L}$ have been checked, merge collinear segments.



Incremental

Algorithm 3: *Incremental*

1. Start by the first 2 points, construct a line
2. Add the next point to the current line model
3. Recompute the line parameters
4. If it satisfies line condition, continue (go to 2)
5. Otherwise, put back the last point, recompute the line parameters, return the line
6. Continue with the next 2 points, go to 2



RANSAC

Algorithm 4: *RANSAC*

1 Initial: A set of N points
2 **repeat**
3 Choose a sample of 2 points uniformly at random
4 Fit a line through the 2 points
5 Compute the distances of other points to the line
6 Construct the inlier set
7 If there are enough inliers, recompute the line parameters, store the line, remove the inliers from the set
8 **until** *Max.N.Iterations reached or too few points left*



Hough Transform

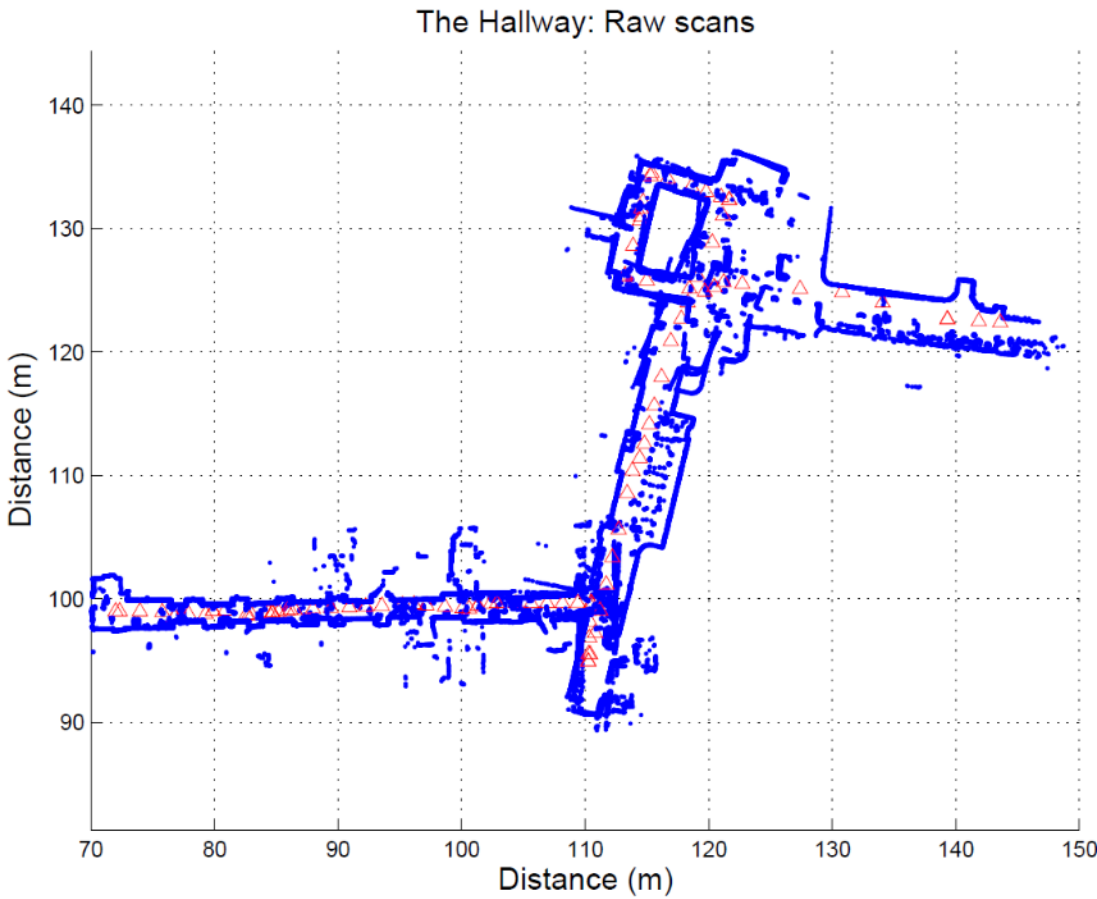
Algorithm 5: *Hough-Transform*

1. Initial: A set of N points
2. Initialize the accumulator array (*model space*)
3. Construct values for the array
4. Choose the element with max. votes V_{max}
5. If V_{max} is less than a threshold, terminate
6. Otherwise, determine the inliers
7. Fit a line through the inliers and store the line
8. Remove the inliers from the set, goto 2



Comparison

- Environment mapped by robot.
- Clustering based on discontinuities.



Comparison

EXPERIMENTAL RESULTS OF THE ALGORITHMS

Algorithm	Complexity	Speed $[Hz]$	N.Lines	Correctness		Precision	
				TruePos [%]	FalsePos [%]	$\sigma_{\Delta r}$ $[cm]$	$\sigma_{\Delta \alpha}$ $[deg]$
Split-Merge + Clus.	$N \times logN$	1470	641	86.0	8.9	1.95	0.74
Incremental	$S \times N^2$	344	561	77.8	5.9	2.04	0.72
Incremental + Clus.		617	567	79.2	5.1	2.04	0.76
Line Regression	$N \times N_f$	364	577	76.4	10.1	1.99	0.80
LR + Clus.		384	562	75.8	8.4	1.97	0.79
RANSAC	$S \times N \times N.Trials$	29	749	75.6	31.5	1.68	0.77
RANSAC + Clus.		93	547	70.7	12.2	1.37	0.70
Hough Transform	$S \times N \times NC + S \times NR \times NC$	8	825	82.0	32.5	1.63	0.76
HT + Clus.		9	600	79.5	10.0	1.51	0.67
EM	$S \times N1 \times N2 \times N$	0.6	1153	78.6	53.7	2.09	0.97
EM + Clus.		0.7	709	80.3	23.1	1.58	0.73



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Localization

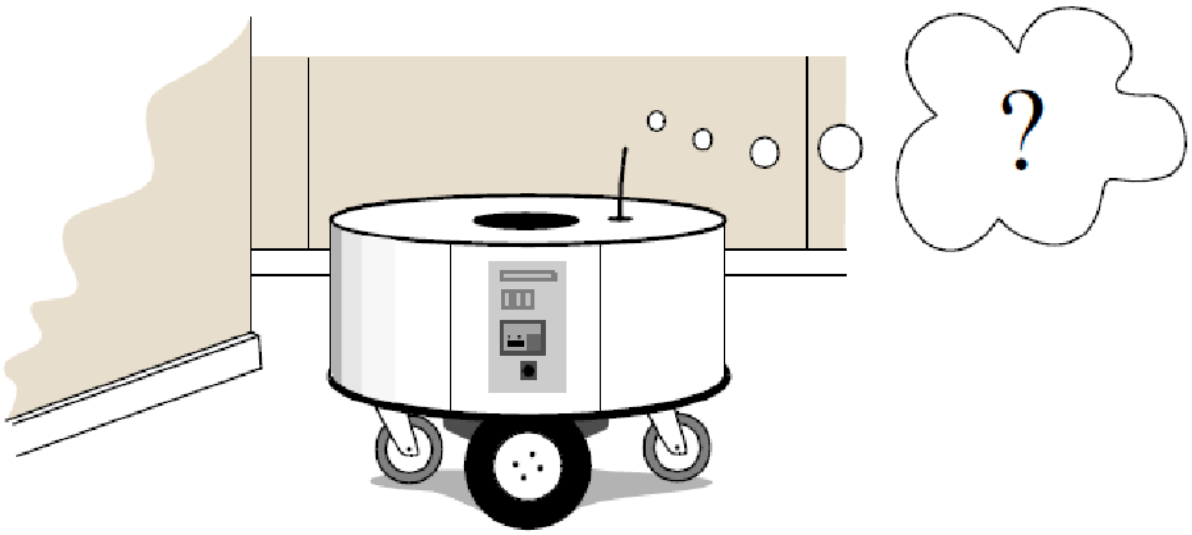


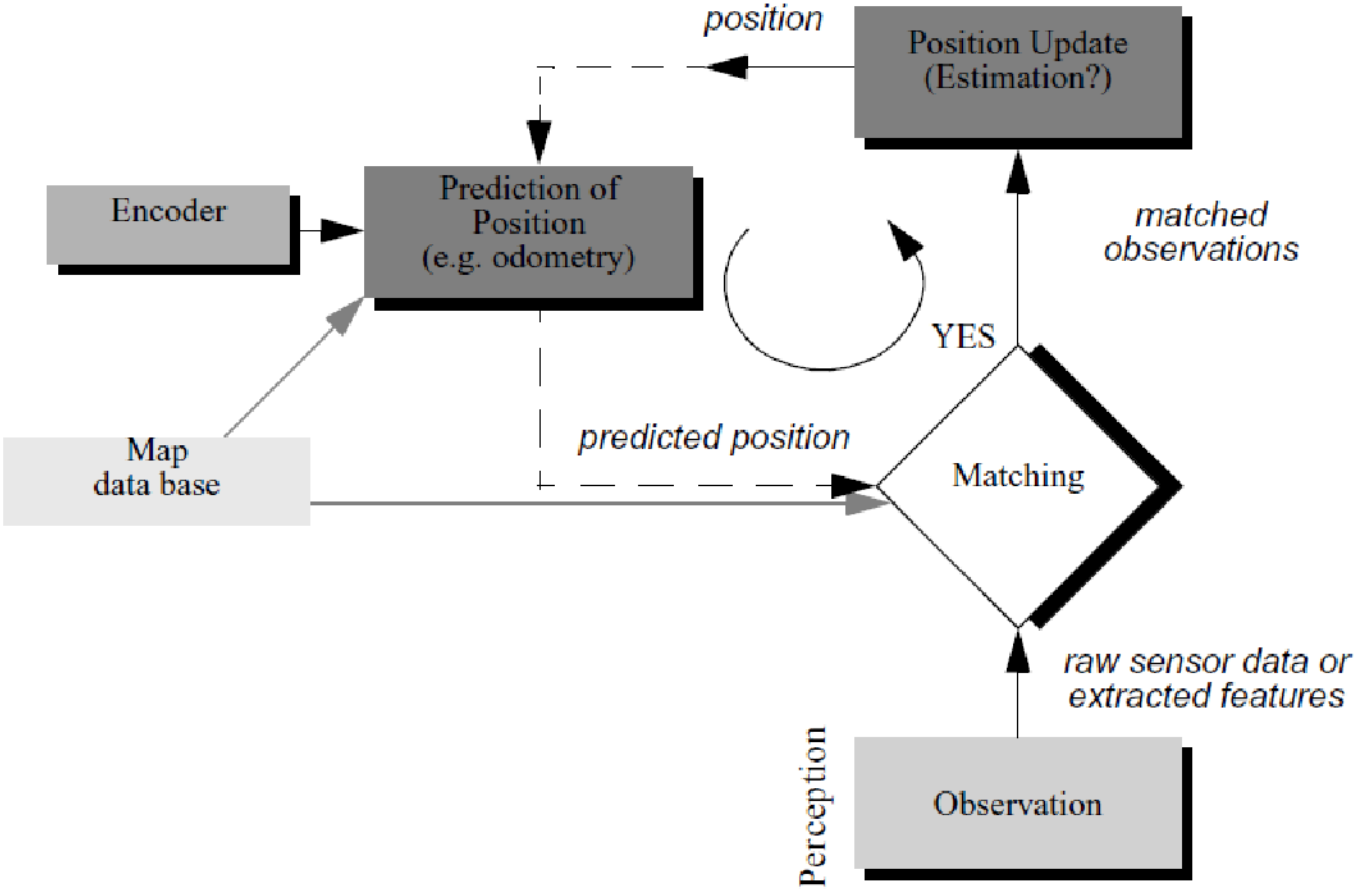
Figure 5.1
Where am I?

Localization

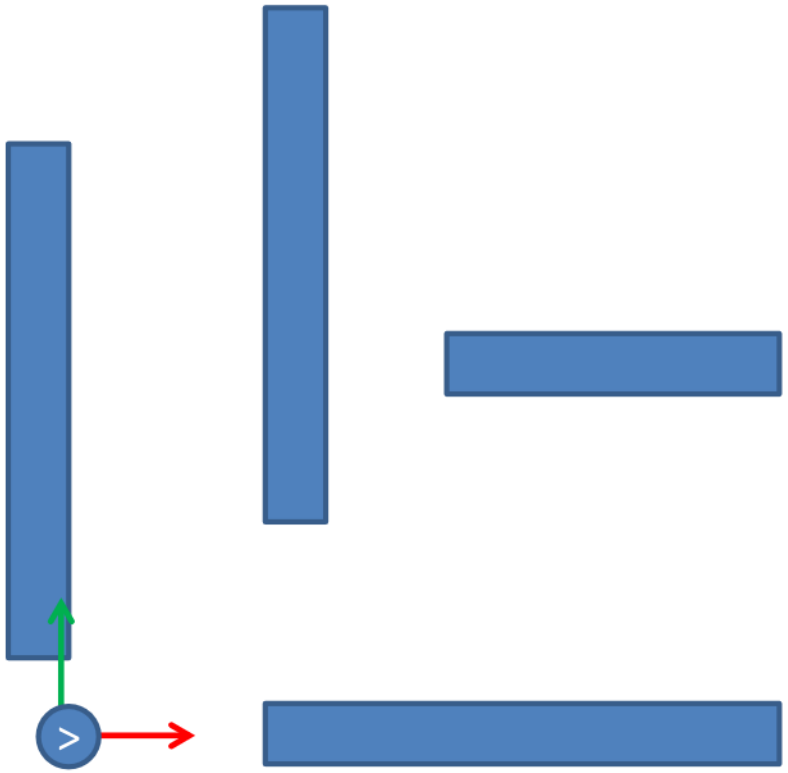
- Two cases
  - When we have a map of the environment (Localization)
  - When we need to construct the map as we go along (Simultaneous Localization And Map building (SLAM)).

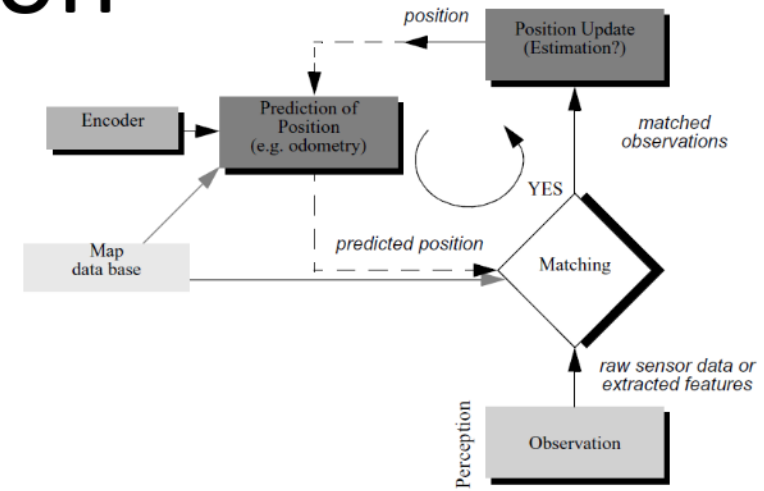


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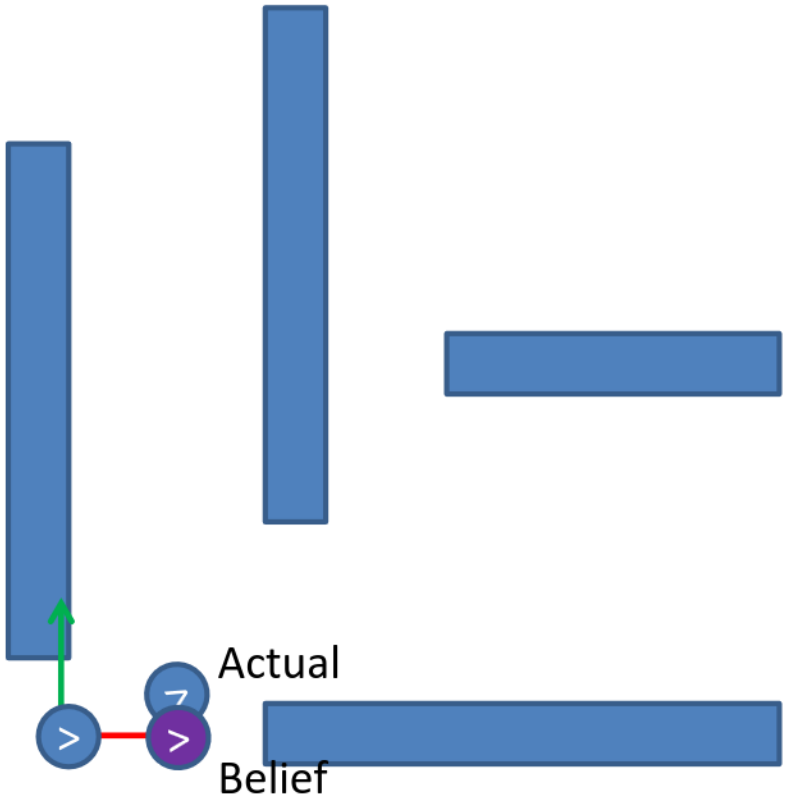


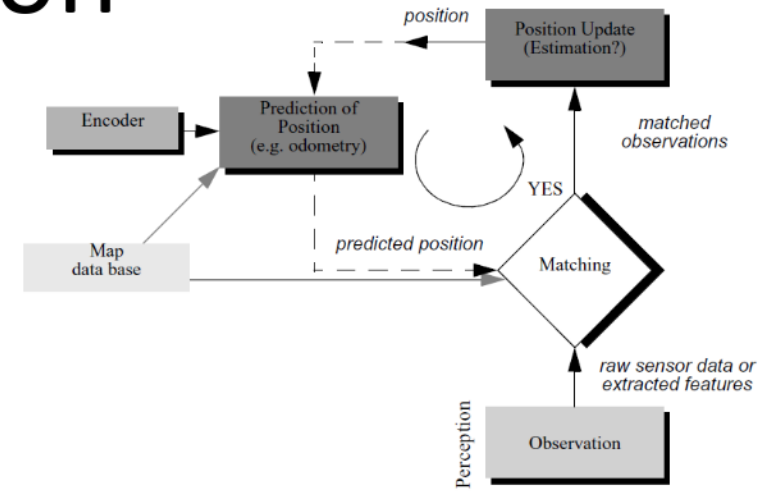
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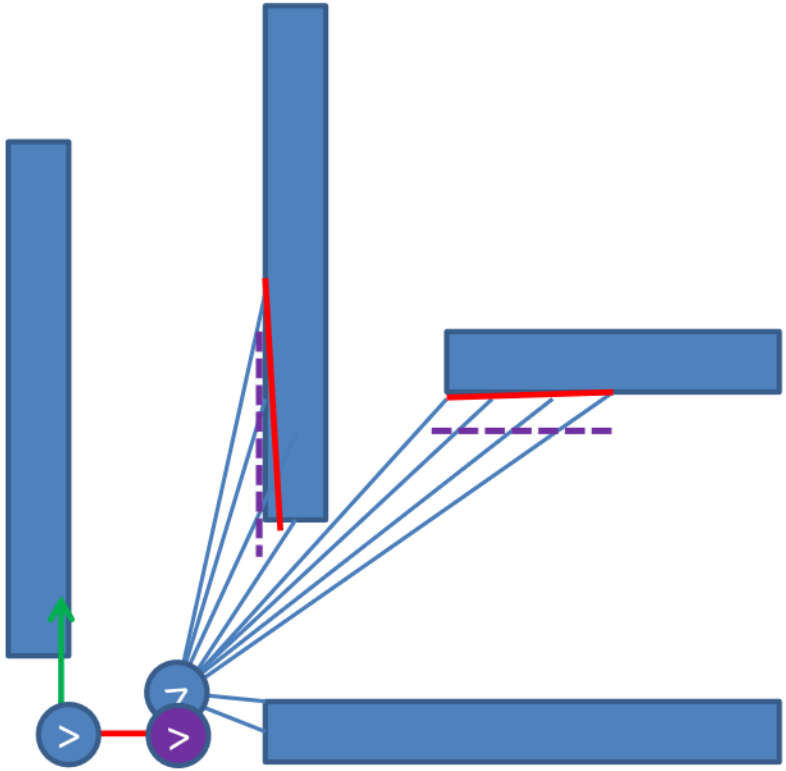


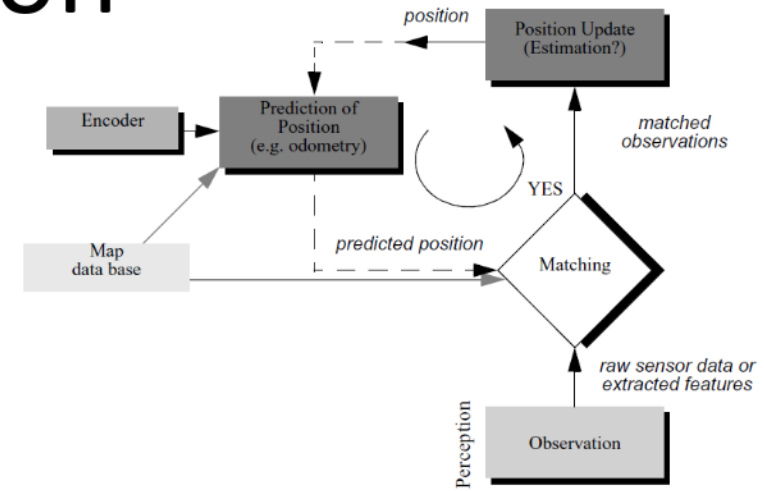
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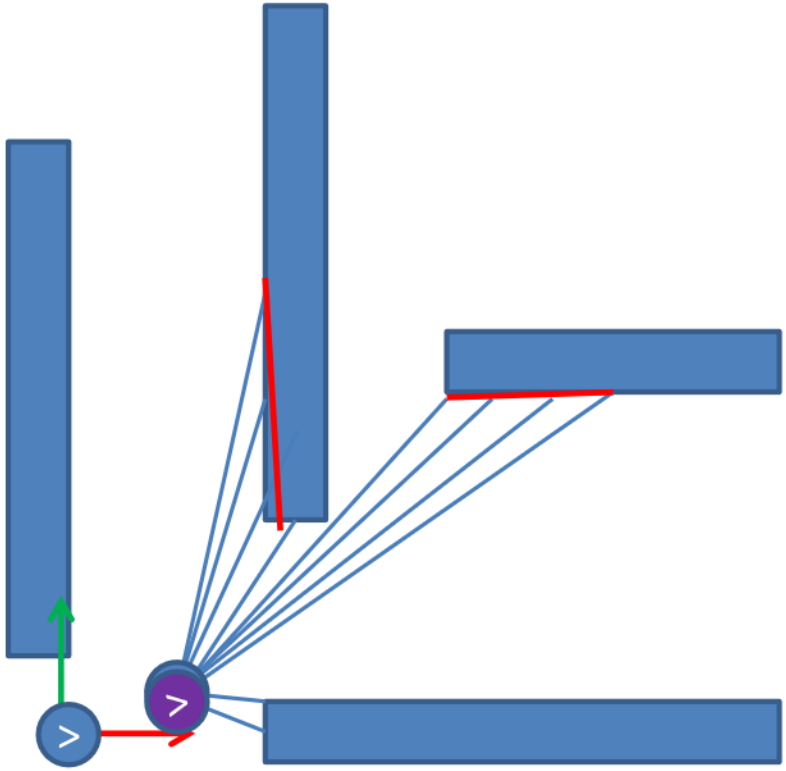


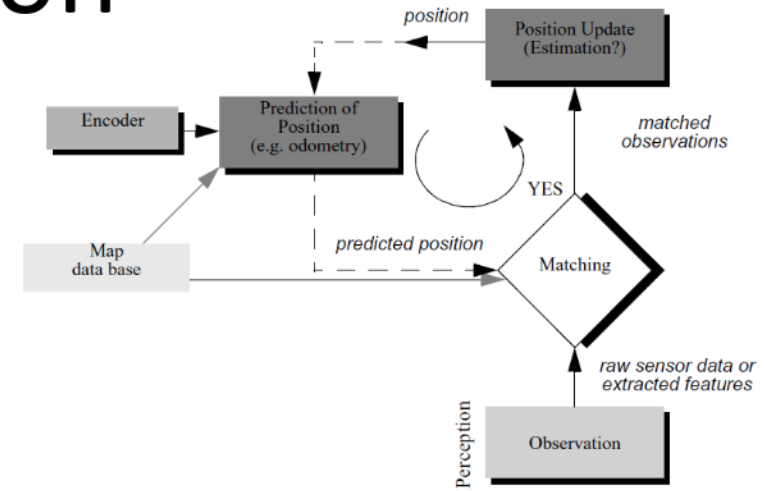
Localization





Localization





Localization

- Two general challenges – noise and aliasing
  - Sensor noise – what to trust
    - More involved features such as lines are more robust than single points.
  - Sensor aliasing – non-unique sensor readings
    - The more complex the feature, the more unique a matching/aliasing we can make
  - Effector noise – odometry errors, accumulate



Summary

- Feature extraction
  - Features
  - Line Representation and Fitting
  - Line Extraction Algorithms
- Introduction to Localization

